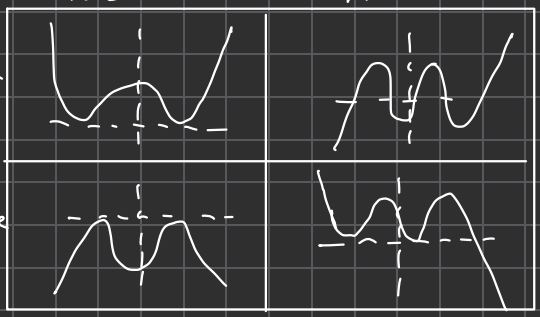


7.1 Polynomial expressions:
A polynomial is an expression composed of the form ax^n, where $n \in \mathbb{Z}^+$.

Polynomial Curves
All you need for a sketch is the roots, y-intercept, and shape (dependent on coefficient of the highest order term).

An order of a polynomial is the highest index present with a non-0 coefficient.
A polynomial of order n has n-1 turning points.

Turning points are when gradients change from $>0 \Rightarrow <0$, or vice versa



This can be used to find $\lim_{x \to \infty} f(x)$ or $\lim_{x \to -\infty} f(x)$. Even are reflecting in $y=0$, whereas odd have rotational symmetry around the origin (180°).

A constant in a polynomial shows a value of 'y' when $x = 0$.
Factorised polynomials can be used to find roots

7.2 Dividing Polynomials
Result of dividing a polynomial is a quotient.
Expression we divide by is the divisor

If the divisor goes into the expression exactly - it is a factor, otherwise the division will produce a remainder also.

7.3 Polynomial Equations:

- Factor Theorem:

If $(x-a)$ is a factor, $f(a)=0$, and $x=a$ is a root of the equation $f(x)=0$.

If $f(a) \neq 0$, then $(x-a)$ is not a factor - as there is a non-0 remainder

- Remainder Theorem:

if $f(a) \neq 0$, $f(a)$ is the remainder of the division

$\frac{f(x)}{(x-a)} \left(= q(x) \cdot (x-a) + f(a) \right)$

Division can be done by inspection or long division

$$\frac{2x^3 - 5x^2 - 6x + 9}{(x-1)} = 2x^2 - 3x - 9$$

Create the largest order, and multiply down for inspection

$$x-1 \;\overline{)\; 2x^3 - 5x^2 - 6x + 9} \;\Rightarrow 2x^2 - 3x - 9.$$

$2x^3 - 2x^2$

$-3x^2 - 6x$

$-3x^2 + 3x$

$-9x + 9$

$-9x + 9$

0

Divide the term, remultiply by divisor, subtract from dividend.